

RUBIC'S CUBE

CATEGORIES

2*2 CUBE

3*3 CUBE

4*4 CUBE

5*5 CUBE

BIG CUBES (6×6 AND LARGER) – REDUCTION METHOD

PYRAMINX

MEGAMINX

SKEWB

MAPLE LEAF SKEWB

POLARIS SKEWB

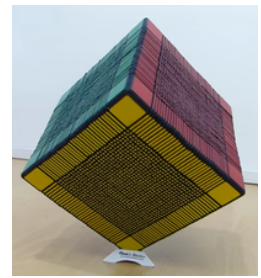
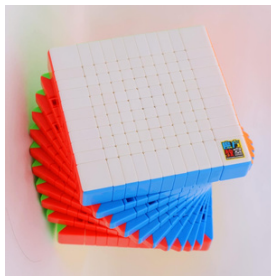
CATEGORIES

There are many different categories within Rubik's cubes. Below are some of the most popular and well-known ones.

I don't want to make this too bloated, so I'll try to keep everything as organized and concise as possible.

Regular NxN Cubes

Most Rubik's Cubes are available up to $11 \times 11 \times 11$, and cubes within this range are easier to find. Higher-order cubes are more rare. Some, like the MoYu $21 \times 21 \times 21$, are commercially made but expensive and uncommon. Even larger cubes, such as $33 \times 33 \times 33$ and above, are usually custom-made or 3D-printed rather than mass-produced.



Shape Mods

Shape mods are cubes with modified cuts, shapes, or turning mechanisms.

Availability varies by design:

- **Fisher Cube** – Available from $3 \times 3 \times 3$ up to higher orders like $7 \times 7 \times 7$ (limited availability).
- **Axis Cube** – Commonly available as $3 \times 3 \times 3$; higher-order versions are rare and mostly custom-made.
- **Windmill Cube** – Mostly available as $3 \times 3 \times 3$, with some higher-order variants being uncommon.
- **Mastermorphix** – Mainly available in $2 \times 2 \times 2$, $3 \times 3 \times 3$, and some larger variants.
 - Mastermorphix, Mastermorphix 4×4 , Gigamorphix
- **Ghost Cube** – A Ghost Cube is a shape mod of the 3×3 cube where the pieces are heavily skewed and offset, making the solved state hard to recognize. Unlike normal cubes, it must be solved by shape and piece orientation rather than just colors. It is known as one of the more difficult shape mods because even a small misalignment can make the cube look unsolved. Mostly available as $3 \times 3 \times 3$, with limited higher-order versions.

Many shape mods also have mirror variants, such as Mirror Fisher, Mirror Windmill, and Mirror Axis.

Other Common Shape Mods

These are additional shape-mod puzzles with unique designs and mechanisms:

- **Lens Cube** – A rounded shape mod with curved cuts.
- **Rhombohedron** – A skewed cube-shaped puzzle with rhomboid faces.
- **Octahedron Cube** – A cube mechanism in an octahedral shape.
- **Cross Cube** – A cube with extended cross-like sections.
- **Heart Cube** – A cube shaped like a heart.
- **Clover Cube Mod** – A clover-shaped modification with unique piece arrangements.
- **Maze Cube** – A cube with maze-like patterns or paths on its faces.
- **Siamese Cube** – Two or more cubes joined together, sharing pieces.
- **Hollow/Void Cube** – A cube with open or empty sections.
- **Transparent Cube** – A clear-bodied cube where internal mechanisms are visible.



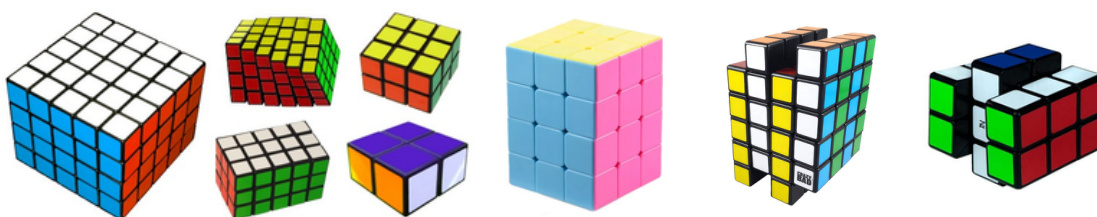
Shape mods are not limited to cubes ($N \times N \times N$); they can also be made from cuboids ($N \times M \times L$).

Cuboids

Cuboids are puzzles with $N \times M \times L$ dimensions, where the three dimensions represent the number of layers along each axis. Unlike cubes, the dimensions do not have to be equal. Two dimensions can be the same ($N \times N \times M$), or all three can be different ($N \times M \times L$).

Examples: $1 \times 1 \times 2$, $1 \times 1 \times 3$, $1 \times 2 \times 2$, $1 \times 2 \times 3$, $2 \times 2 \times 3$, $2 \times 2 \times 4$, $2 \times 3 \times 3$, $2 \times 3 \times 4$, $3 \times 3 \times 1$, $3 \times 3 \times 2$, $3 \times 3 \times 4$, $3 \times 3 \times 5$, $3 \times 4 \times 4$, $3 \times 4 \times 5$, $4 \times 4 \times 2$, $4 \times 4 \times 3$, $4 \times 5 \times 5$, $5 \times 5 \times 3$ and many more possible $N \times M \times L$ combinations.

- The most popular cuboid is the $3 \times 3 \times 1$, commonly known as the Floppy Cube, which also exists in other variants such as the fidget spinner-style cuboid. Other popular cuboids include the $2 \times 2 \times 3$, $2 \times 3 \times 3$, $3 \times 3 \times 2$, and $3 \times 3 \times 4$.



Supercubes / Picture Cubes (Require Center Orientation)

Supercubes and picture cubes add center orientation as an extra challenge. Unlike regular $N \times N$ cubes where centers only need the correct color position, these cubes require each center piece to be rotated correctly as well. A cube may look solved by color but still be incorrect if logos, symbols, or images on the centers are misaligned.



Mirror Series

Mirror cubes are shape-mod versions where the pieces are distinguished by size and shape rather than color. Mirror cubes are commonly available from $2 \times 2 \times 2$ up to $7 \times 7 \times 7$, with sizes up to $5 \times 5 \times 5$ being easier to find. Higher-order versions are rarer and are often custom-made or 3D-printed through specialty sites.

Some cubes also have mirror variants of other shape mods, such as Mirror Fisher Cube, Mirror Windmill Cube, and other mirror-style designs.



Gear Cubes

Gear cubes use interlocking gears that cause neighboring pieces to rotate together. A single turn rotates both the face and the visible gears, creating movement unlike a standard Rubik's Cube.

While the classic Gear Cube is based on a 3×3 mechanism, many variants exist with additional cuts or altered shapes.



- Gear Cube (3×3 Gear Cube), Gear Ball, Gear Cube Extreme, Gear Cube Ultimate, Gear Mixup, Gear Barrel, Gear Cube Plus, Gear Pyraminx, Gear Megaminx, Gear Cube shape mods and custom gear puzzles.

Skewb Puzzles (Corner-Turning Puzzles)

Skewb puzzles turn around their corners instead of their faces. Each move rotates one corner along with adjacent pieces, giving them a unique solving style that differs from face-turning cubes. The Skewb mechanism has inspired numerous shape modifications and higher-complexity variants.

- Skewb, Skewb Diamond, Skewb Ultimate, Master Skewb, Elite Skewb, Skewb Xtreme, Fisher Skewb, Skewb Cube, Skewb Mixup, Skewb shape mods (Barrel, Rhomboid, Windmill, etc.), Picture and Super Skewbs (require center orientation where applicable)



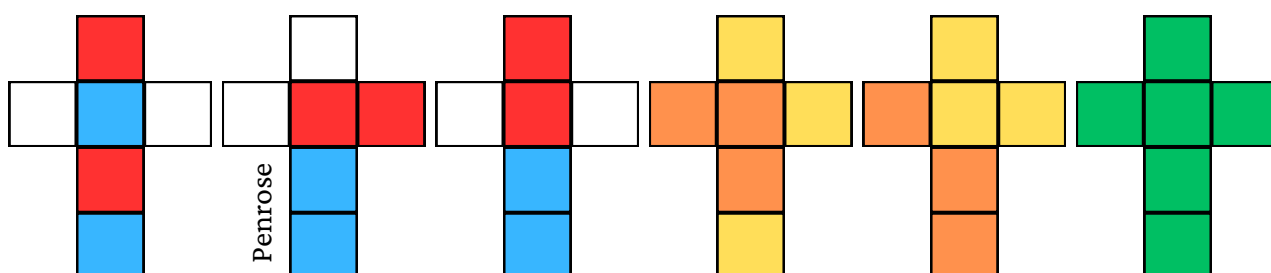
- Pyraminx Family and Megaminx Family are analogous to the standard NxN cube family, but on different polyhedra (tetrahedron and dodecahedron, respectively). Like NxN cubes, they have progressively larger and more complex versions.
- Square-1 Family (including Square-0, Square-1, Square-2, etc.) and Puppet Cubes are shape-shifting puzzles with unique mechanisms that differ significantly from standard face-turning cubes.
- Redi, Ivy, Dino, Rex, and Curvy Copter are popular alternative puzzle families. Most are simpler than a 3×3 to learn but often have several variants, shape mods, and higher-complexity versions.
- Other notable puzzle families include Morpfix, Mixup, Latch, Gear, Skewb, Clock, Cuboids, Crazy Cubes, Void Cubes, and Bandaged Puzzles, each introducing different turning mechanisms or solving concepts.
- There are also hybrid (combo) puzzles that combine features from multiple families. Examples include Mirror Fisher Cube, Gear Mixup Cube, Gear Pyraminx, Crazy Skewb, Void Skewb, Mirror Windmill Cube, Gear Megaminx, and many custom or 3D-printed combinations. These hybrids often require techniques from more than one puzzle family.

These are the most common twisty puzzle families and variations. Beyond these, there are countless custom designs, shape mods, hybrids, and limited-edition puzzles. New combinations are created regularly, so covering every variation would be impractical.

My Current Collection

- Standard cubes: 2×2, 3×3, 4×4, 5×5
- Polyhedra: Pyraminx, Megaminx
- Gear: Gear Cube
- Mirror: 3×3 Mirror Cube
- Shape mods: Fisher Cube, Axis Cube, Windmill Cube, Mastermorphix
- Shape-shifter: Square-1
- Other puzzles: Magic Pentacle Cube, Maple Leaf Cube, Polaris Cube
- Custom mods: Six cubes modified into custom color schemes, including Penrose, Force Cube, and several other experimental color combinations.

Six cubes modified



2*2 CUBE

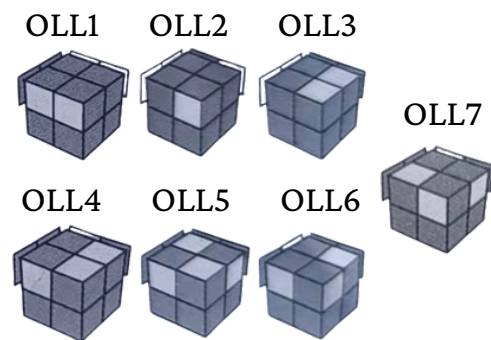
Step 1: Solve the Down Face

Choose your favorite color as down face and match up all these colors (there are 4 pieces in this color), you just need to solve it same as the corner pieces of a 3x3 cube.

Step 2: OLL - Solve the Top Face

In this step you need to orient the color of the top face. The result should be that the opposite color to the bottom color will be completed. There are 7 cases.

- **OLL1:** $R^2 U^2 R U^2 R^2$
- **OLL2:** $F R U R' U' F'$
- **OLL3:** $F R U R' U' R U R' U' F'$
- **OLL4:** $R U R' U R U^2 R' F R U R' U' F'$
- **OLL5:** $F R U R' U' F'$
- **OLL6:** $U R U^2 R' U' R U' R'$
- **OLL7:** $R U R' U R U^2 R'$

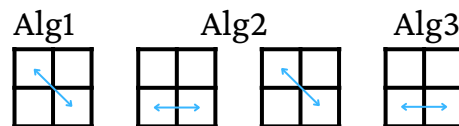


Step 3: Permute Remaining Colors

At last we need to match up all colors of the top and bottom layers, there are 5 cases.

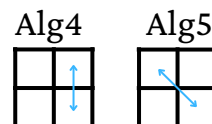
If the top layer and bottom layer are not solved, you need to learn these 3 algorithms.

- **Alg 1:** $R^2 B^2 R^2$
- **Alg 2:** $R' U R' B^2 R U' R'$
- **Alg 3:** $R^2 U' R^2 U^2 F^2 U R^2$

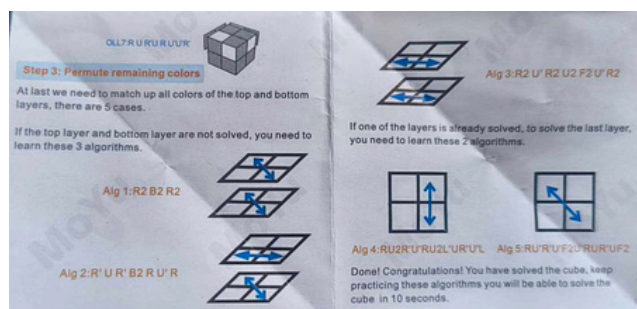
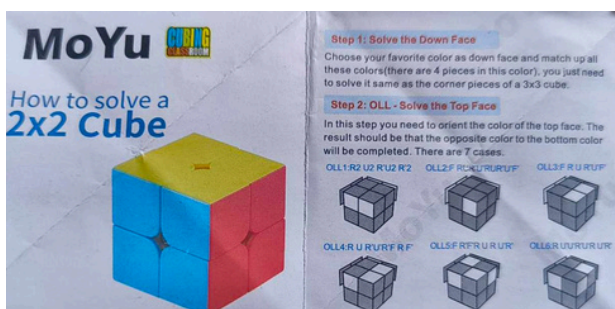


If one of the layers is already solved, to solve the last layer, you need to learn these 2 algorithms.

- **Alg 4:** $R U^2 R' U' R U^2 L' U R' U' L$
- **Alg 5:** $R U R' F' R U R' U' R' F R^2 U' R' U'$



Done! Congratulations! You have solved the cube, keep practicing these algorithms you will be able to solve the cube in 10 seconds.



3*3 RUBICS CUBE

This is a cheatsheet for returning cubers—not a step-by-step beginner tutorial. Use these LBL (Layer By Layer) algorithms for fast reference. Each "code" below is an algorithm in standard Rubik's notation.

1. First Layer (White Face)

Solve the white cross by aligning white edge pieces with matching centers. Then insert white corners into the bottom layer by positioning each white corner between its matching edge pieces and using intuitive turns to place it correctly.

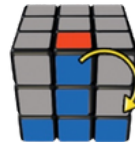
2. Second Layer (Middle Edges)

Keep the solved white face on the bottom during this step.

Find a top-layer edge with no yellow. Match its front color to center, then insert it:

If the edge goes to the left:

$U' L' U L U F U' F'$



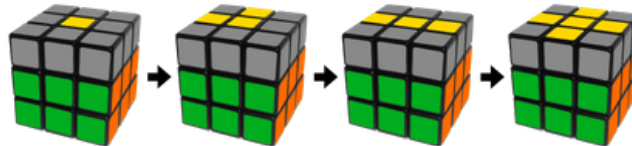
If the edge goes to the right:

$U R U' R' U' F' U F$

Repeat until all four middle-layer edges are in place.

3. Yellow Cross (OLL – Edge Orientation)

Form the yellow cross on the top face using: $F R U R' U' F'$



4. Yellow Corners Orientation (OLL – Corner Orientation)

Turn all yellow corners so yellow stickers face up using: $R U R' U R U^2 R'$

Repeat until all yellow stickers are on top.

5. Corner Positioning (PLL – Corner Permutation)

Move yellow corners into their correct locations (ignore orientation).

$U R U' L' U R' U' L$



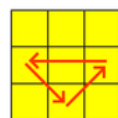
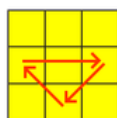
Repeat until all corners are in the right spots. If two are solved, put them in the back and do the move. If none are solved, do it once, then check again.

6. Edge Positioning (PLL – Edge Permutation)

Finish the cube by cycling the last layer's edges. Use the correct version based on the direction of the edge cycle.

Clockwise cycle

$R^2 U R U' R' U' R' U' R' U R'$



Counterclockwise cycle

$R' U R' U' R' U' R' U R U R^2$

Apply as needed. The cube should now be fully solved.

4*4 RUBICS CUBE

Cube Notation

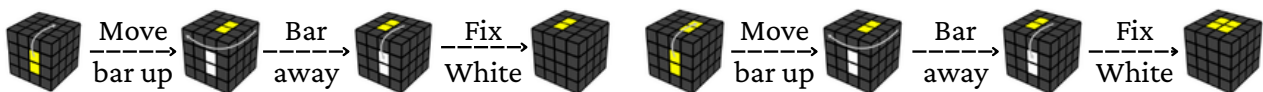
- R, L, U, D, F, B – Turn outer Right, Left, Up, Down, Front, Back face Clockwise
- R', L', U', D', F', B' – Turn outer face Anti-clockwise
- r, l, u, d, f, b – Turn inner slice behind Right, Left, Up, Down, Front, Back
 - also notated as 2R, 2L, 2U, 2D, 2F, 2B
- Rw, Lw, Uw, Dw, Fw, Bw – Turn two layers: outer + inner (wide move)
- R2, L2, U2, D2, F2, B2 – Turn outer face 180°
- x, y, z – Rotate whole cube: x = rotate like R, y = rotate like U, z = rotate like F

Step 1 – White Center

- Find two white center pieces.
- Align them to form a 2-piece white bar on one face. Make another white bar and combine both using inner slices to create the full white 2×2 center.
- After every move, restore the outer layers so you don't break existing bars.

Step 2 – Yellow Center (Opposite White)

- Build one yellow bar using two yellow centerpieces. Build a second yellow bar.
- Combine both to complete the yellow center opposite the white center.



Step 3 – Remaining Four Centers

Hold the cube with white on the left and yellow on the right.

The colors should stack around the cube in this order going upwards:

Blue → Orange → Green → Red (BOGR).

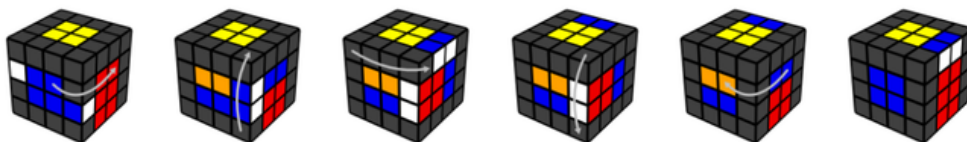
Note: Some cubes might have this order reversed (downwards), like mine.

To solve, first form bars on each face, then line them up correctly, and finally connect them using inner slice turns.

Step 4 – Pair the Edges (Reduction Stage)

Each 4×4 edge has two “wing” pieces that must be paired.

Algorithm for Pairing and Flipping: **$RUR'FR'F'R$**



- Hold two matching edge pieces on the same layer (e.g., both red-white edges).
- Use slice moves (turn inner layers) to bring them together.
- Apply the above algorithm to pair or flip them.
- After each pair, undo the slice so your centers remain solved.

- Continue this for 10 edges while not scrambling the paired edges.

Step 5 – Last Two Edge Pairs

- Position both unsolved edge pairs in the same layer.
- Use the same slice-flip algorithm again ($R U R' F R' F' R$).
- Slice, Swap and re-slice to restore your centers.



- Now, all 12 edge pairs will be complete.

Step 6 – Solve Like a 3×3 Cube

Once centers and edges are reduced:

- Treat each 2×2 center block as a single center.
- Treat each paired edge as one edge.
- Solve using any 3×3 method (Layer-by-Layer, CFOP, etc.).

Step 7 – Fix Parity Errors

Unique 4×4 cases (cannot happen on 3×3):

(a) OLL Parity – Flipped Edge Pair

When two opposite top-layer edges are flipped incorrectly.

Algorithm: $r U2 x r U2 r U2 r' U2 l U2 r' U2 r U2 r' U2$

- Hold the cube with flipped edges on top and facing you.
- Executes a double edge flip correction.

(b) PLL Parity – Two Edges Swapped

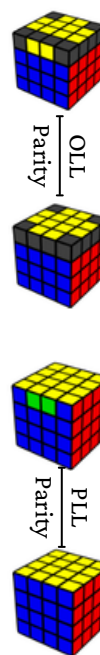
When two opposite edge pairs are swapped.

Algorithm: $r2 U2 r2 u2 r2 u2$

- Hold the cube with swapped edges front and back.
- This swaps the two edges and completes the cube.

Step 8 – Final Adjustments

After fixing parities, complete your last layer just as on a normal 3×3.



Fixing Center Piece Errors

Do note that your cube may use a different color scheme.

Check a corner piece to determine correct center positions.

If centers are misplaced,

- To swap U and F centers:
 - $Rw' F2 Rw2 U2 Rw'$
- To swap U and D centers:
 - $Rw2 U2 D2 Rw2$

5*5 RUBICS CUBE

Step 1 – Solve the Centers (Each 3×3 center block)

- Solve edge pieces of the center: Create center bars, bringing the correct center pieces together using slice turns.
- Solve corner pieces of the center: Attach corner center pieces to complete the 3×3 block. Fully solve the white center first. Solve the yellow center opposite white.
- Solve the remaining four side centers, working systematically.

JPerm's clever layer-shift pattern to insert and fix center pieces on 5×5

1. Position the target center piece on the top layer above its slot (e.g., front center).
2. Perform a wide move on the left or right slices to push the piece down:
 - If moving from the left side, do **Lw** or from the right side, do **Rw'**.
3. Rotate the front face (F) clockwise or counterclockwise depending on the needed orientation.
4. Move just the affected inner slice (not wide) down one layer:
5. Rotate front face opposite to Step 3 orientation.
6. Undo the wide move on the left or right to restore layers:
 - Undo by **Lw'** or **Rw** depending on initial step (Step 2).
7. Repeat the Step 3 front face turn, then move the inner front slice back up to complete the insertion.

Step 2 – Pair the Edges

- Pair the three edge pieces (wing edges) for each edge.
- Use slice moves to bring matchable pieces together.
- When edge pieces don't align properly, use the edge flip algorithm:
 - **R U R' F R' F' R**
- Store paired edges in the top or bottom layer. Undo slice moves to restore centers after pairing. Use special maneuvers if edges are difficult to pair.

Step 3 – Solve Like a 3×3 Cube

- After pairing all edges and completing centers, the 5×5 effectively becomes a 3×3.
- Solve it using your favorite 3×3 method (CFOP, Roux, beginner method, etc.).

Step 4 – 5×5 Parity Fix

- You may encounter edge parity errors unique to 5×5.
- Use the parity-fixing algorithm:
 - **Rw U2 x Rw U2 Rw U2 Rw' U2 Lw U2 3Rw' U2 Rw U2 Rw' U2 Rw'**
- This algorithm swaps two edge pieces that are incorrectly paired.
- After parity correction, continue as usual to finish the solve like a 3×3.

BIG CUBES (6×6 AND LARGER) – REDUCTION METHOD

The Reduction Method is the standard approach for solving all larger NxN cubes. The goal is to reduce the puzzle into a virtual 3×3 by first solving the centres and pairing the edges. Once this is complete, the cube can be solved exactly like a normal 3×3.

Step 1 – Solve the Centres (All 6 centres)

- A common order is: Solve **Two opposite centres** → **Two adjacent centres** → **Remaining two centres**.
- Build the centres by creating bars and combining them into complete centre blocks. As the cube size increases, centre building becomes more dependent on efficient slice moves and, occasionally, commutators for difficult cases.

	Fixed center	Remember colour scheme	OLL parity	PLL parity
Even Cubes	✗ No	✓ Yes	✓ Yes	✓ Yes
Odd Cubes	✓ Yes	✗ No	✗ No	✓ Yes

Solve Last Two Centers

Referenced from [How to Solve Last Two Centers on Big Cubes](#) | [Cube Ed](#)

• Phase 1: The Swap

Align: Position your target centers in the **U** (Top) and **F** (Front) faces, aligned in the same vertical column (ideally on the right side).

Engage Column A: Move the **first column** (containing your target piece) **up** once.

Rotate: Move the **U** layer **anti-clockwise (U')**.

Engage Column B: Move the **second column** (the one that now contains the missing spot) **up** once to complete the initial swap.

• Phase 2: The Restoration

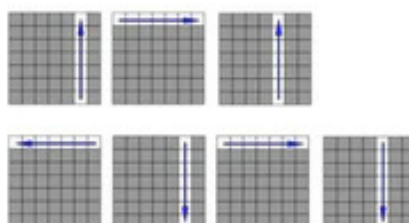
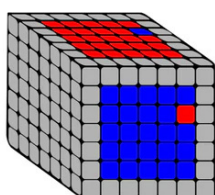
Rotate Back: Move the **U** layer **clockwise (U)** to realign the rows.

Restore Column A: Move the **first column down** once.

Rotate Again: Move the **U** layer **anti-clockwise (U')** to align for the final restoration.

Restore Column B: Move the **second column down** once to finish the sequence.

EXAMPLE
(slightly diff.)



Step 2 — Pair the Edges

Once all centers are solved, pair every edge until each behaves as one large edge.

General process:

1. Find matching edge pieces.
2. Bring them together using inner slice moves.
3. Store solved edges in the top or bottom layer.
4. Repeat until every edge is paired.

This is exactly the same method used on the 4×4 and 5×5, just with more edge pieces.

Step 3 — Solve Like a 3×3

Once the centers and edges are complete, the cube is effectively reduced to a 3×3.

Simply solve it using your preferred method:

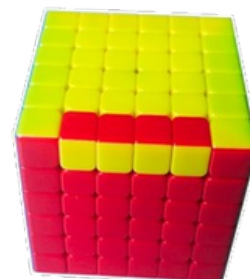
- Cross, F2L, OLL, PLL

Step 4 — Fix Parity (When Needed)

- After reducing the cube to a 3×3, you may encounter parity—cases that are impossible on a standard 3×3. These only occur because of the way larger cubes are constructed.
- The parity algorithms are the same ones used on the 4×4 and 5×5. On larger cubes, simply turn the appropriate number of inner layers.

OLL Parity (Even Cubes Only)

- Occurs on: 6×6, 8×8, 10×10, ... - single flipped last-layer edge, creating a case that cannot exist on a normal 3×3.
- **How to solve it:** Use the same OLL parity algorithm as the 4×4, but turn more inner layers.
- **Ex:** 4×4 → Turn 2 layers, 6×6 → Turn 3 layers, 8×8 → Turn 4 layers, so on..



6×6 : $3Rw\ U2\ x\ 3Rw\ U2\ 3Rw\ U2\ 3Rw'\ U2\ 3Lw\ U2\ 3Rw'\ U2\ 3Rw\ U2\ 3Rw'\ U2\ 3Rw'$

8×8 : $4Rw\ U2\ x\ 4Rw\ U2\ 4Rw\ U2\ 4Rw'\ U2\ 4Lw\ U2\ 4Rw'\ U2\ 4Rw\ U2\ 4Rw'\ U2\ 4Rw'..$

Edge Parity (All cubes 5×5 and larger)

- During edge pairing, one edge cannot be completed normally because two matching edge pieces are incorrectly swapped or flipped.
- Use the same edge parity algorithm from your 5×5 guide. For larger cubes, perform the algorithm while turning the correct number of inner layers for your cube size.

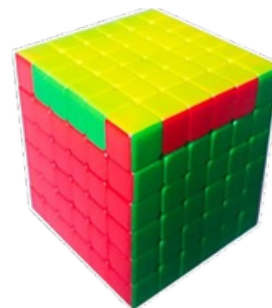


Here: $2R'\ U2\ 2L\ F2\ 2L'\ F2\ 2R2\ U2\ 2R\ U2\ 2R'\ U2\ F2\ 2R2\ F2$

2L/2R represent the second inner left/right layers.

PLL Parity (Even cubes Only)

- After completing OLL, you might find PLL where find two edges swapped—a case that cannot occur on a standard 3×3.
- Hold the cube with your last layer on top, then perform the same PLL parity algorithm used on the 4×4, turning the appropriate number of inner layers.



6×6 Algorithm: ***3Uw2 3Rw2 U2 3Rw2 R2 U2 3Rw2 3Uw2***

8*8 Algorithm: ***4Uw2 4Rw2 U2 4Rw2 R2 U2 4Rw2 4Uw2***

GEAR CUBE

Notation: Standard cube notation: U, D, L, R, F, B (Up, Down, Left, Right, Front, Back). Because of the Gear Cube's structure, only double turns are possible

- **$R = R2$; $R2 = R4$ and such.**

Step 1: Solve the Corners

- This is intuitive — solved using only R, D, F moves (max 4 moves needed).
- The more you solve, the better you become at recognizing patterns, making it more effective than trying to learn every variant.

Step 2: Solve the Centers

- After Step 1, at least 2 of 6 centers are usually already solved.
- Turn the layer containing the misaligned centers until they line up with their adjacent side colors.

Step 3: Permute the Edges (Gears)

- Find a band where one edge color already matches the opposite face (e.g., the green–white edge in the blue–white edge). Hold the cube with that band on top.
- Apply: $R2, U, R2, U$ (equivalent to $R4 U2 R4 U2$ in standard cube notation).
- Rotate the cube to the next unsolved band and repeat until all edge pieces are in the correct positions.

Step 4: Orient the Edges

- Some edges may be in the right spot but twisted wrong.
- Simply rotate the layer along the band until each edge piece is facing the correct way — no algorithm needed, just turning until they align.

Step 5: Solve the Inner Edges (only on some Gear Cube versions)

- Skip this step if your cube doesn't have inner edge pieces.
- Hold the cube so two inner edges of the same color face up.
- Apply: **$(R, U) \times 3$**

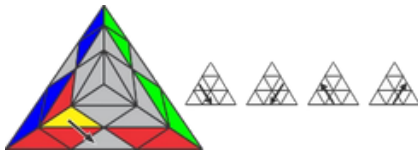
PYRAMINX

Step 1: Align Tips and Centers

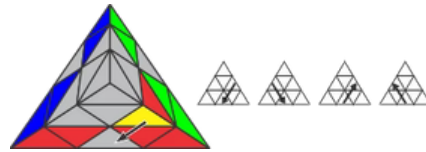
- Turn each tip to match adjacent center colors.
- Choose a base color (e.g., yellow).
- Rotate centers so all 3 matching centers are on one face.

Step 2: Solve Base Layer Edges

- Find each edge with the base color and Bring it above its correct position.
- Insert using:



Left insert: $L R' L' R$



Right insert: $R' L R L'$

- Insert a random piece incase edge you are looking for is in the bottom layer.

Step 3: Solve Top Layer Edges

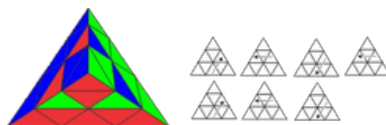
Clockwise:

$R U R' U R U R'$



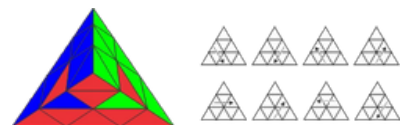
Anti-clockwise:

$R U' R' U' R U' R'$



Swap 2 adjacent edges:

$L R' L' R U R U' R'$



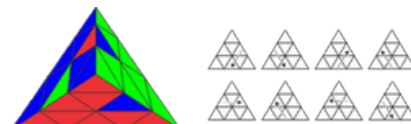
- If you know how to swap two adjacent edges and rotate colors (clockwise or counterclockwise), that's enough to solve the cube. For speed-solving, learn both methods and the algorithms below.
- First, identify the single edge pair with both colors pointing outward (e.g. blue and green respectively in below)—this signals a swap + rotation combo. Then check the back edge: if it can be solved with a clockwise rotation, use the clockwise algorithm; if not, use the counterclockwise one.

Rotate 3 edges and flip 2 edges:

Clockwise: $L R' L' R U R U' R'$



Anti-clockwise: $R' L R L' U L' U' L$



MEGAMINX

Except for the last layer, the rest of the puzzle can be solved by applying familiar 3×3 cube methods:

- Solve one face and its adjacent edge pieces.
- Complete the first layer.
- Solve the middle layers using standard edge-insertion algorithms.
- Continue building layers around the puzzle until only the final layer remains.



Edges

Left: $F' R' F'^2 R F$
Right: $F L F^2 L' F'$



Top/Last Layer star

1st case : $F U R U' R' F'$
2nd case : $F R U R' U' F'$



Top/Last Layer Cycle edges

Clockwise: $R U R' U R U^3 R' U$
AntiClockwise: $U' R U^2 R' U' R U' R'$



Top/Last Layer Cycle edges

Clockwise: $L' U^2 R U'^2 - L U^2 R' U'^2$
Anticlockwise: $R U'^2 L' U^2 - R' U'^2 L U^2$



Orient corners

Doing U rotations bring the unsolved corners to the highlighted position one by one. Repeat the $R' D' R D$ algorithm for each piece until they are all positioned correctly.

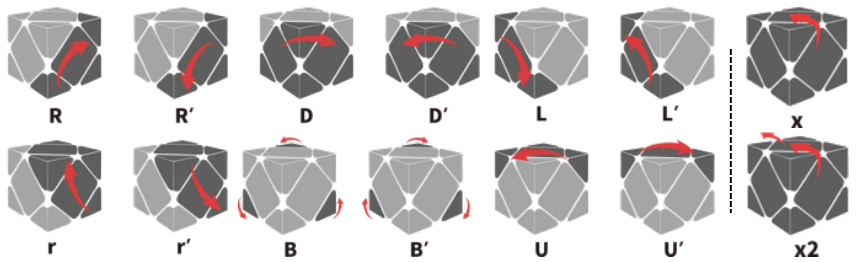
- Use the JPerm tutorial for visual learning:
 - <https://youtu.be/oVRooYDvRqg>

Most of the solving is intuitive and depends on how well you've understood 3×3 concepts, along with experience and pattern recognition. These methods aren't exact or exhaustive—I've come across a few slightly trickier cases and additional insights over time.

That said, as a beginner-friendly guide or a quick memory refresher, this tutorial is really helpful.

SKEWB

Cube Notation



Solving Steps



Step 1: Solve the White Corners

White corners are mostly solved intuitively.

Placement: White center on top.

Solve all four white corners using the appropriate algorithm for each case.

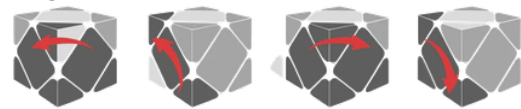
Case A: White sticker on the front-left

- Algorithm: $D R D' R'$



Case B: White sticker on the front-right

- Algorithm: $D' L' D L$



Step 2: Solve the Yellow Corners

Placement: White face on the front-left.

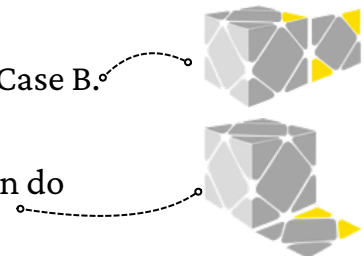
Algorithm “S”: $R' r R r'$. Key algorithm used to solve the rest of the Skewb.

Case A: Two yellow stickers are on the right-back face.

- Position the cube as shown and do Algorithm S to reach Case B.

Case B: No yellow stickers are on the right-back face.

- Place the two adjacent yellow stickers at the bottom, then do Algorithm S to solve all yellow corners.



Step 3: Solve the Yellow Face

Placement: White face on the front-left, yellow center on the left-back.

Align the yellow center with the yellow corners using:

Algorithm: $S x2 S$ i.e. $R' r R r' x2 R' r R r'$

I understood the algorithm to work as a up-up-down-down method which helps to better understand it.

Step 4: Solve the Centers (Finish the Skewb)

Observe the solved faces and match one of the cases below.

Case A: Only the white and yellow faces are solved.

Placement: White and yellow on top and bottom (either way).

- Do $S x2 S$ once to reach Case B.



4 Unsolved Centers

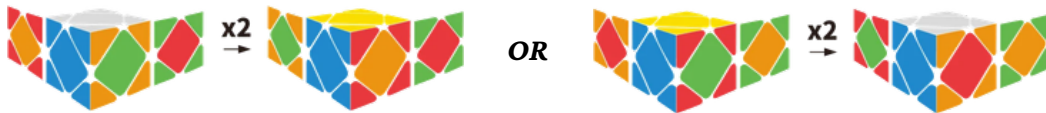
SKEWB

Case B: White, yellow, and one side face are solved.

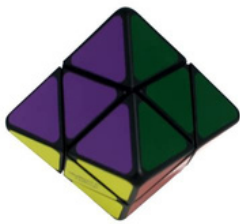
Placement: Solved side on the front-left, white and yellow on top and bottom.

- If the three unsolved centers cycle clockwise, keep the position.
- Otherwise(Counter clockwise) , do $x2$ to swap the white and yellow faces.

Then do $S \ x2 \ S$ once to solve the Skewb.



Skewb Diamond



Diamond Octahedron



Master Skewb



Skewb Ultimate

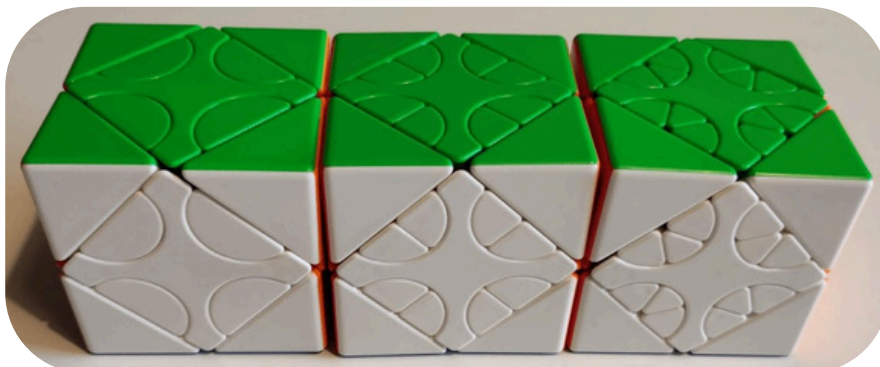


Curvy Skewb



**Mastered the Skewb?
Explore these related
puzzles.**

Skewb Xtreme



Skewb Mixup I , II , III

MAPLE LEAF SKEWB

A Maple Leaf Cube is basically a Skewb with two extra piece types per face: 4 big triangles and 4 leaf pieces, all built around a fixed center with a color indicator. First, solve the triangles and leaf pieces to restore the puzzle to a solvable shape, then solve it just like a standard Skewb.

- The **first essential move** is the **180° flip**: turn a center layer to the right, move it down to the bottom, then bring it back up to the front. This rotates the square on that face by 180° without disturbing any pieces you've already solved.
 - ie To move from front to right side using **D** and so on..
 - To move from bottom to front using **L'**.
- The **second essential move** is the **swap turn**: bring a piece next to its target with a partial turn, rotate the corresponding edge or corner section by 180° to swap the two pieces, then undo the partial turn to restore everything else.

Phase 1 — Fix the Big Triangles (centers pair up)

Goal: get all 4 big triangles on each face matched with that face's center color.

1. Pick a color, find a triangle piece of that color.
2. Bring it to the front face, with top piece being intended color.
3. Do a partial turn (Either right-up or left-up as needed)
 - a. If it lines up with a wrong-colored triangle on that face → do the swap turn to bring it in.
 - b. If it doesn't line up → do the 180 flip on that face's center layer first (this rotates the whole square), then it will line up — now do the swap turn.
4. Repeat this for all four triangles of one color, then move on to the next color. The order of the faces doesn't matter, but leave the face opposite the completed one until the end.
5. **Tricky case:** if your target piece keeps landing only on faces that are already finished, no matter which 180 flip you try — move it to an unfinished adjacent face first (one swap turn), then flip from there. This "**Saves/Store**" it somewhere useful before you bring it when needed.
6. **Last two faces:** you only need to actively fix one of them (e.g. white) — fixing it automatically fixes its adjacent (eg. green). If a piece won't line up directly:
 - Use an already-solved face (e.g. orange) as a temporary helper.
 - Swap your problem piece onto that helper face (swap turn).
 - Bring helper+piece to the front, do the 180 flip, then swap the piece into its real home (swap turn).

- Undo the flip on the helper face by reversing it with bottom → right (instead of right → bottom) so the helper face lands back exactly where it started.
- Bring the helper piece back home.

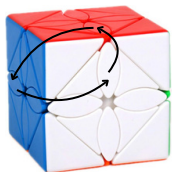
When all 6 faces show 4 correctly matched big triangles — Phase 1 done.

Phase 2 — Fix the Leaf Pieces

Goal: same idea, but leaf pieces swap in 3-cycles (three pieces rotate positions at once), so you set up a piece on the left face as your "helper."

Basic 3-cycle move - (Commutator*)

1. Get the leaf piece you want to fix pointing at the top/front face, in the slot that will swap it home (use the 180 flip if it's not lined up yet).
2. Put a face that is not yet finished on your left side (this is your helper/sacrifice face — never use an already-completed Leaf here).



This move cycles the three pieces counter-clockwise. The entire phase is solved by repeatedly using this cycle, with simple setup moves to position the correct pieces before each cycle.

3. Bring the front section up (partial turn) → 180° turn → undo partial turn. This drops your target leaf piece into place.
4. Bring the left section up (partial turn) → 180° turn → undo partial turn. This pulls the leaf piece you just placed safely out of the way, preserving it.
5. Bring the front piece (now holding the piece that was swapped out in step 3) back up to its home face → swap turn. Fixes another triangle/leaf.
6. Bring the left section back up one more time → swap turn. This restores the piece from step 4 to its real home.
7. Net result: three pieces have cycled into their correct spots at once.

Special case A — target piece would land on an already-finished slot

- If simply rotating a face 180° would dump your "extra" piece onto a finished section, first relocate a block of pieces off that target face (swap it out to some other face, flip that face into the helper slot, then bring it back) so there's room to land the 3-cycle without clobbering finished work. Track your setup move so you can undo it exactly afterward.

Special case B — piece needs to jump to a totally different face

- If a piece can't be aimed at its target with a single flip, first move it (via a swap turn) onto an intermediate face, then do the 180 flip from there to aim it correctly, then run the 3-cycle as normal.

Golden rule for all of Phase 2: whatever face you place on the left during the 3-cycle must NOT already be solved — you're temporarily "spending" one of its correct slots

to route pieces through. Always undo your setup turns afterward so nothing ends up actually broken.

- Work through leaf pieces face by face; order doesn't matter. When all faces show correct triangles + leaves around each center, every "big square" is a solved unit and the cube is now just a scrambled Skewb.

Phase 3 — Solve Like a Normal Skewb

Since each face's center/triangles/leaves now move as one locked block, solve it with your normal Skewb method:

1. Solve one layer of 4 corners (e.g. white face corners).
2. Solve the opposite 4 corners (e.g. yellow), using up/down/up/down type corner algorithms.
3. Position the remaining centers/side pieces into their correct spots (rotate top layer, cycle the 3 remaining centers as needed).
4. Done — fully solved Maple Leaf Cube.

Abit more detailed info in [Skewb](#) for phase 3

HELPFUL TIMESTAMPS

1. Solving the Large Triangular Pieces (0:50 - 13:25)

- This is the initial phase where you pair the triangular pieces with their respective center indicators.
- **Special Setup Case:** You will encounter situations where pieces do not align for a simple 180-degree swap (10:50 - 13:25). To fix this, use a setup move to relocate the piece to an adjacent face, rotate the center, perform the swap, and then reverse the setup moves to restore the faces already completed.

2. Solving the Leaf Pieces (13:25 - 35:45)

- This process uses a three-cycle method to position the leaf pieces.
 - Leaf Pieces (Part 1): (13:25 - 24:45) Starting Pieces.
 - Leaf Pieces (Part 2 - Remaining Pieces): (24:45 - 31:30) Ending pieces
- **Tricky Special Case:** Only 2 Pieces remaining
 - Pieces Pointing Toward Each Other (28:00 - 31:30): Perform a setup move to relocate a group of pieces to a non-critical slot (like the bottom face). This creates a neutral position to execute your three-cycle swap, then reverse the setup moves to restore your progress.
 - Pieces Pointing Elsewhere (32:10 - 35:45): If a piece isn't aligned to the target face, move it to an adjacent face. Use a 180-degree center rotation to reorient it, perform the three-cycle to insert the piece, and systematically undo your setup maneuvers to preserve the rest of the solve.

3. Final Skewb Phase (35:45 - 37:37)

- Once the leaf pieces and large triangles are finished, the puzzle is solved like a standard Skewb by aligning the corners and completing the final orientation of the centers.

POLARIS SKEWB

The Polaris Cube is a shape-shifting twisty puzzle with diagonal turning layers.

Unlike a standard Rubik's Cube, a single turn usually leaves the puzzle misaligned, so all movements require two turns to return it to an aligned shape.

The puzzle consists of two types of pieces:

- **Big pieces** – the pointed corner pieces.
- **Small pieces** – the pieces between the corners.

Notes:

- Big and small pieces must always alternate (Big → Small → Big → Small).
- If two big pieces are next to each other, the puzzle is misaligned.
- Never force a turn. If something feels stuck, realign the puzzle before continuing.

The solve can be broken into five phases:

- Restore the shape → Solve the white corner pieces → Solve the centers → Solve the small pieces → Fix the final layer.

Phase 1 – Restore the Shape

Before solving colors, restore the cube to its normal shape.

1. Turn the corner points until they all face one another.
2. Continue making diagonal turns until every face is flat and aligned.
3. Remember that most moves require two turns to restore alignment.
4. If a move feels blocked, undo it and try another section instead of forcing it.

Phase 2 – Solve the White Corner Pieces

1. Choose the white center as your starting face.
2. Locate each white big (corner) piece.
3. Move each white corner around the white center using diagonal turns.
4. Repeat until all four white corners surround the white center.
5. Rotate the cube about 60° and confirm each corner matches its adjacent side color. Ignore the small pieces for now.

Phase 3 – Solve the Centers

Once the white corners are complete, solve the centers. This phase uses Skewb center algorithms—it is not enough to simply place yellow opposite white.

1. Keep the solved white corners intact.
2. Position the centers so that:
White is opposite to Yellow, Blue is opposite to Green, Red is opposite to Orange
3. Use the required Skewb center algorithms to cycle the centers into their correct positions.

4. Continue until every center matches the surrounding corner colors.

Layer Shift Algorithm: This algorithm is used repeatedly while positioning pieces without disturbing solved sections.

1. Turn the right side down → Turn the left side down → Turn the working section twice clockwise → Reverse the moves by bringing the right side up and then the left side up. (Skewb Algorithm)

Repeat as needed until the required center or corner is correctly positioned.

Phase 4 – Solve the Small Pieces

Before starting, make sure all corner points are facing outward.

Work one section at a time, beginning with the white face.

Case A – One Small Piece Is Already Correct

1. Find a corner section where one small piece is already in its correct position.
2. Hold the puzzle so that this correct piece stays in the stationary (non-moving) section while you work.
3. Place the next small piece into the remaining open slot temporarily, even if it is not its final position.
4. Turn the working section over.
5. Bring the correct small piece into its final position using Up → Turn → Down.
6. Repeat until the entire corner section is solved.

Case B – No Small Pieces Are Correct

1. Find the two small pieces that belong in the current corner section.
2. Position one of them correctly.
3. Move the other piece temporarily into the stationary section so it stays out of the way.
4. Restore the layer.
5. Bring the final required piece into place using Up → Turn → Down.
6. Repeat until the corner section is complete.

Repeat until all four sections around the white face are solved, then repeat the same process for the yellow face without disturbing the completed white side.

Phase 5 – Final Layer (Parity)

- Occasionally, the final four small pieces belong to the correct section but are rotated incorrectly. Hold the puzzle diagonally and repeat the following sequence until the pieces resolve:
- Up → Turn → Down → Turn → Up → Turn → Down →.. Until corner is solved.

Check the puzzle after each repetition and repeat if necessary.

- Once all colors and pieces are matched: Turn every section back into its flat, aligned position (not diagonal/staggered).